INCH-POUND

MIL-C-63296 (AR)
AMENDMENT 5
25 March 1992
SUPERSEDING
AMENDMENT 4
3 April 1991

MILITARY SPECIFICATION

CASE, CARTRIDGE, 105MM, M115B1

This amendment forms a part of Military Specification MIL-C-63296 (AR), dated 22 January 1980, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1

STANDARDS MILITARY: Add:

"MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-410 - Non-destructive Testing Personnel Qualification and Certification."

DRAWINGS ARMY: Add:

"9280433 - Envelope Drawing, 105MM Cartridge Case, Ultrasonic Inspection.
9280434 - Longitudinal Calibration Standards, 105MM Cartridge Case, Ultrasonic Inspection.
9280435 - Circumferential Calibration Standard, 105MM, Cartridge Case, Ultrasonic Inspection."

PAGE 4

3.8.4: Delete "No cartridge case shall have a burr." and substitute "Burrs shall be limited to those described in 105mm cartridge case 'Visual Defect Standards'."

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PAGE 5

4.3.1:

Change "254*" to "262".

Delete "* All samples must meet the requirements for thermaltreatment and high stress expansion test as performed in the uncoated state."

PAGE 6

Table I First article inspection:

Delete in its entirety "Cartridge Case - Prior to Coating" and substitute:

"Cartridge Case - Prior to Coating

Ultrasonic inspection	262 (Note 2)	3.8.1	4.5.7.1".

Cartridge Case - After Coating:

Delete "254" and substitute "262".

Add:

"Tension test (Info only)	4	3.3	4.5.4
Macro examination	4	3.8.2	4.5.5
Hardness	20	3.3	4.5.3".

Add Note 2:

"Note 2. All units must be ultrasonically inspected after performing high stress expansion test. All non-compliance units shall be removed. Evidence of cracks prior to or after tapering shall be cause for rejection of case."

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4.4.2.1: Delete in its entirety and substitute new paragraph 4.4.2.1 (included herein as Page 3).

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4.4.2.2: Delete in its entirety and substitute new paragraph 4.4.2.2 (included herein as Pages 4 through 6).

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Case, Cartridge - Prior to Coating				10520214
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	High stress expansion test		100%(a)	3.1	4.5.2
102	Ultrasonic inspection		100%	3.8.1	4.5.7.1

NOTES: (a) All non-compliance units must be removed from the lot. Evidence of cracks in any portion of the case shall be cause for rejection.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Case, Cartridge - After Coating	1 OF 3	10520214
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cracks		100%(a)	3.8.1	4.5.7
102	Metal defective		100%(a)	3.8.1	4.5.7
103	Thickness of coating (See 4.5.6)			3.4	4.5.6.1
104	Salt spray (See 4.5.6)			3.1	4.5.6.2
105	Maximum profile and alignment		100%(a)	3.5	4.5.8
106	Excess pressure ballistic test (See 4.4.3.1.2)			3.6	4.5.9.2.1
107	True position of pitch diameter of thread with body		0.40%	3.1	Gage
108	True position of primer hole counterbore with pitch diameter of thread		0.40%	3.1	Gage
109	True position of primer hole thread relief with pitch diameter of thread		0.40%	3.1	Gage
110	Pitch diameter of thread, max.		0.40%	3.1	Gage
111	Minor diameter of thread, max.		0.40%	3.1	Gage
112	Length total, max.		0.40%	3.1	Gage
113	Diameter of primer hole counterbore		0.40%	3.1	Gage
114	Depth of primer hole counterbore, min.		0.40%	3.1	Gage
115	Length from flange to bottom of primer hole counterbore, max.		0.40%	3.1	Gage
116	Depth of thread relief		0.40%	3.1	Gage
117	Protective coating defective		0.40%	3.4	Visual
118	Thermal treatment		100%(a)	3.3.1	4.5.1

NOTES: (a) All non-compliance units must be removed from the lot.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.2	Case, Cartridge - After Coating		10520214
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
119	Hardness	4	(b)	3.3	4.5.3
120	Macro-examination	4	(b)	3.8.2	4.5.5
121	Diameter of flange, min.		0.40%	3.1	Gage
122	Diameter of mouth		0.40%	3.1	Gage
123	Length of mouth diameter, min.		0.40%	3.1	Gage
124	Inside diameter before taper at mouth end		0.40%	3.1	Gage
125	True position of inside diameter before taper at mouth end with mouth diameter		0.40%	3.1	Gage
126	Thickness of flange, min.		0.40%	3.1	Gage
127	Thickness of base, min.		0.40%	3.1	Gage
128	Wall thickness at first datum from base		0.40%	3.1	Gage
129	Wall thickness at second datum from base		0.40%	3.1	Gage
130	Wall thickness at third datum from base		0.40%	3.1	Gage
131	Wall thickness at mouth		0.40%	3.1	Gage
132	Contour of inside of base and sidewall not smooth		0.40%	3.1	Visual
133	Ultrasonic acceptance stamp missing		100%	3.1	Visual
Minor					
201	Pitch diameter of thread, min.		0.65%	3.1	Gage
202	Diameter of primer hole thread relief		0.65%	3.1	Gage
203	Thread damaged or not full		0.65%	3.8.3	Visual
204	Body dented or damaged		0.65%	3.1	Visual
205	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES: (b) Reject on one defect.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Case, Cartridge - After Coating	3 OF 3	10520214
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
206	Wax coating missing		0.65%	3.1	Visual
207	Evidence of poor workmanship		0.65%	3.8.4 & 3.8.5	Visual
208	Diameter of body under flange, min.		0.65%	3.1	Gage
209	Location of first basic diameter from flange, min.		0.65%	3.1	Gage
210	Location of second basic diameter from flange, min.		0.65%	3.1	Gage
211	Location of third basic diameter from flange, min.		0.65%	3.1	Gage
212	Location of fourth basic diameter from flange, min.		0.65%	3.1	Gage
213	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
214	Surface finish improper		0.65%	3.1	Visual
215	Length total, min.		0.65%	3.1	Gage
Special					
301	Tension test (Information only)	(c)		3.3	4.5.4

NOTES: (c) Samples used for the hardness test shall be used in the tension test (see 4.5.4).

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4.4.2.3: Major defect: delete "None defined".

Add: "101 Cracks and other metal defects (cartridge cases) just prior to packing into carton 100% 3.8.1 Visual".

PAGE 14

* 4.4.3.1.2:

After "Production lot" add: "(Cases not ultrasonically tested)".

Line 6, delete: "...provided for test." and substitute "...tested."

* Add:

"4.4.3.1.3 Production lot (Cases ultrasonically tested) Twenty five cartridge cases from each lot (see 4.4.1), shall be ballistically tested in the excess pressure phase. Results of the test shall be evaluated in accordance with the production acceptance criteria set forth in Table III, except for all defect characteristics the acceptance criteria are accept on 0 defects, reject on 1 defect."

PAGE 15

4.5.4: Delete in its entirety and substitute:

"4.5.4 Tension test (Information only). Four cartridge cases from each lot shall be subjected to this test. The test specimens shall be taken from each cartridge case in accordance with the standard position and size of the tensile specimen chart described in Figure 1. The yield strength, tensile strength and elongation shall be determined in accordance with the procedures prescribed by ASTM-E8. The results shall be recorded for information only."

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4.5.6.1: Last sentence, delete "...rejection of the lot." and substitute "...testing additional samples in accordance with MIL-STD-105, Inspection Level II, 0.40% AQL."

Delete 4.5.7 and 4.5.7.1 and substitute:

"4.5.7 Cartridge case cracks and flaw detection. The cartridge case shall be visually examined for workmanship and cracks and shall comply with the requirements of paragraph 3.8 or established visual standards. The visual standards shall be mutually agreed upon between the government and the contractor. All rejected cases shall be set aside and held for government inspection.

4.5.7.1 Ultrasonic inspection. Each cartridge case after final machining, prior to coating, shall be inspected for metal defects using as reject standards drawings 9280434 and 9280435 and equipment satisfying the requirements of drawing 9280433. Deviation from standards must be approved by the respective technical agency. All cartridge cases accepted by the ultrasonic equipment shall be stamped with a "U". Cartridge cases rejected by the equipment shall be stamped with an "R" and the rejected cases subsequently removed. The stamps utilized shall not have sharp corners which will create corrosion. The stamping shall comply with the requirements as stated in cartridge case drawing 10520214. The quantities of cartridge cases inspected and rejected shall be counted and logged for determination of the reject rate."

Add paragraphs 4.5.7.1.1, 4.5.7.1.2, 4.5.7.1.3 and 4.5.7.2:

"4.5.7.1.1 Calibration procedures. The ultrasonic inspection equipment, specified by drawing 9280433, shall be calibrated, at a minimum, at the beginning, middle and end of an eight hour work shift. If for any reason the reject standards are not rejected by any of the channels, the contractor must re-calibrate the equipment and re-inspect all accepted cartridge cases ultrasonically inspected since the last acceptable calibration.

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4.5.7.1.2 Rejection rate - ultrasonic equipment. When the ultrasonic inspection equipment rejection rate exceeds 5.0 percent for the period between calibrations, the contractor shall check equipment calibration. If the calibration is acceptable, all cartridge cases previously accepted by the equipment during this period shall be re-inspected. During re-inspection, if the reject rate exceeds 1.0 percent, all cartridge cases scheduled for re-inspection or accepted by re-inspection shall be placed aside until the cause for the excessive reject rate is determined. When the cause is determined, disposition of the held cartridge cases shall be at the discretion of the contracting officer. At any time, if the contractor proves that an excessive reject rate is equipment related, he may apply to the contracting officer for re-inspection of those rejected cartridge cases found during this time.

4.5.7.1.3 Ultrasonic equipment - operator qualification. The operators of the contractor's equipment for ultrasonic inspections required by this specification shall be certified and qualified "Level I". Also, any set-up, calibration or interpretation personnel shall be certified and qualified "Level II". The qualification and certification criteria shall be as prescribed by MIL-STD-410 and other applicable specifications.

4.5.7.2 Visual inspection. Four power (4x) magnification shall be used in the examination of the exterior surface of the sidewall for a minimum distance of eight inches (203.20 mm) forward of the base (flange). All other surfaces may be examined without magnification."

NOTE: The margins of this Amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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